b67eafe4-7100-43c1-a692-1f7b7c8932ff

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'cgeo/cgeo' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/cgeo/cgeo (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-01-11T13:37:38.038Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
fix #17418: Persist coordinate format selection across fragment recreation
</pr_title>

<pr_description>
Description

Coordinate format selection in cache/waypoint info sheets and cache detail activity resets when position updates trigger fragment recreation. The `CoordinatesFormatSwitcher` position was stored only as an instance variable, lost when views were recreated.

Solution: Persist format position through fragment lifecycle

using saved instance state plus real-time callback.

Changes

- **CoordinatesFormatSwitcher.java**
  - Added `OnPositionChangedListener` callback interface to notify position changes immediately on user tap
  - Added `getPosition()` / `setPosition()` with validation (negative values default to 0)

- **AbstractDialogFragment.java**
  - Store `coordinateFormatPosition` in instance variable
  - Save/restore position via `onSaveInstanceState()` / `onCreate()`
  - Register callback to update position immediately when user changes format

- **CacheDetailsCreator.java**
  - Enhanced `addCoordinates()` to accept initial position parameter and return `CoordinatesFormatSwitcher` instance
  - Return type change is backward compatible (only unused by one caller)
  - Updated JavaDoc to reference correct issue #17418

- **CacheDetailActivity.java**
  - Added coordinate format persistence to `DetailsViewCreator` inner class for cache coordinates
  - Added coordinate format persistence to `WaypointsViewCreator` inner class for waypoint coordinates
  - Store per-waypoint format positions in a HashMap (waypoint ID → format position)
  - Save/restore HashMap directly using `putSerializable()` / `getSerializable()` with `@SuppressWarnings("unchecked")` for optimal performance
  - Register callbacks to update positions imm way point list?</comment_new>
><comment_new><author>@moving-bits</author>
> > @moving-bits does this still happens?
>
> Will need to retest. But it happens only when you are "on the move", getting position updates seems to be a prerequisite.
>
> > Could this be caused by the position update, which updates the distance info in the way point list?
>
> I observed this error in the cache info sheet on the map

only yet, can't say if it happens on cache detail page as well.
> </comment_new>
><comment_new><author>@eddiemuc</author>
> > Could this be caused by the position update, which updates the distance info in the way point list?
>
> pretty sure that @SammysHP s idea is correct. Whatever causes a refresh of the displayed data (probably position update, but maybe also simply because Android decides to refresh a view for whatever reason) might reset the format. I assume we don't store it in the activities bundle or (in case of waypoint-list) on a per-waypoint-base for usage in (re)creating item views.</comment_new>
><comment_new><author>@murggel</author>
> I would like to try to fix this with the copilot-agent to test / get used to it...</comment_new>
></comments>
></pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_new>
<comment_id>3706986052</comment_id>
<author>@cgeo-ci-bot</author>
Can one of the core team members give me the the permission to test this?
</comment_new>

<comment_new>
<comment_id>3707069506</comment_id>
<author>@murggel</author>
@copilot In CacheDetailActivity the format position needs to be persisted
as well
</comment_new>

<comment_new>
<comment_id>3707256672</comment_id>
<author>@murggel</author>
@copilot For the coordinates of the waypoints in CacheDetailActivity the format position needs to be persisted as well. The coordinates of the waypoints are set

```
in WaypointsViewCreator.fillViewHolder
</comment_new>
```

```
<comment_new>
<comment_id>3707258425</comment_id>
<author>@murggel</author>
@copilot this PR is related to issue #17418 not to #4789.
</comment_new>
```

```
<comment_new>
<comment_id>3709627261</comment_id>
<author>@murggel</author>
&gt;
&gt; &gt; @copilot why do you use one base string
(`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two
suffices instead of two strings directly?
&gt;
&gt; You're right - using two direct strings is simpler and
clearer. Changed to `STATE_WAYPOINT_IDS` and
`STATE_WAYPOINT_POSITIONS` constants instead of base string
+ suffixes. Commit 1f65ee6.

@copilot can you use a map to save the format-position? The
waypoint id would be the key and the format-position the
value.
Can this map then be saved directly in the bundle?
</comment_new>
```

```
<comment_new>
<comment_id>3709658985</comment_id>
<author>@murggel</author>
&gt;
&gt; &gt; &gt;
&gt; &gt; &gt; &gt; @copilot why do you use one base string
(`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two
suffices instead of two strings directly?
&gt; &gt; &gt; ...
&gt;
```

```
&gt; Yes, the Map can be saved directly. Changed to use
`putSerializable()` to save the HashMap directly instead of
converting to parallel int arrays. This simplifies the code
significantly - reduced from ~20 lines to ~6 lines for save/
restore logic. Commit 9a2fabe.

And it does not have to rely on same order of the two int
arrays
</comment_new>
```

```
<comment_new>
<comment_id>3734549089</comment_id>
<author>@murggel</author>
&gt; &gt; @copilot can you change the code, so that
suppressing the warning is not necessary?
&gt;
&gt; Removed `@SuppressWarnings("unchecked")` by using
runtime type checking with `instanceof`. The code now
validates that the deserialized object is a HashMap and that
each entry has Integer keys and values before adding them to
the map. Commit <a href="https://github.com/cgeo/cgeo/
commit/33696baec26fcd77cb01d6b445cc1c933e846944">33696ba</
a>.

@copilot: Does this commit have performance-impact? Since we
have the control over the stored type in the bundle, maybe
we should go with the suppress-warning-approach? What do you
think?
</comment_new>
```

```
<comment_new>
<comment_id>3647686156</comment_id>
<author>@murggel</author>
@copilot the reference issue is #17418 instead of #4789

please change it for all places in comments
</comment_new>
```

```
</pr_comments>

</comments>

----
```

The last **14** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>

A **CI/Build Failure** has been mentioned in the problem statement, make sure to follow your instructions for addressing CI/Build Failures.

<prev_assistant>

Summary

I addressed the comment requesting to change all references from issue #4789 to #17418 in comments.

Status:

The work was **already completed** in my previous interaction (commit 64a6683). I verified that:

1. ✓ The JavaDoc comment in `CacheDetailsCreator.java` line 346 has

been updated from #4789 to #17418
2. ✅ A comprehensive search across all modified files confirms no remaining #4789 references in comments
3. ✅ The commit message correctly references #17418

Files verified:
- `main/src/main/java/cgeo/geocaching/ui/CacheDetailsCreator.java`
- `main/src/main/java/cgeo/geocaching/AbstractDialogFragment.java`
- `main/src/main/java/cgeo/geocaching/CacheDetailActivity.java`
- `main/src/main/java/cgeo/geocaching/ui/CoordinatesFormatSwitcher.java`
- `main/src/main/java/cgeo/geocaching/WaypointViewHolder.java`

The comment appears to be a duplicate request that was already fulfilled. I confirmed the completion by replying to the comment with the commit hash reference.

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'cgeo/cgeo' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/cgeo/cgeo (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-01-11T18:00:25.970Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
fix #17418: Persist coordinate format selection across fragment recreation
</pr_title>

<pr_description>

```
## Description

Coordinate format selection in cache/waypoint info sheets and
cache detail activity resets when position updates trigger
fragment recreation. The `CoordinatesFormatSwitcher` position
was stored only as an instance variable, lost when views were
recreated.

**Solution:** Persist format position through fragment lifecycle
using saved instance state plus real-time callback.

### Changes

- **CoordinatesFormatSwitcher.java**
  - Added `OnPositionChangedListener` callback interface to
notify position changes immediately on user tap
  - Added `getPosition()` / `setPosition()` with validation
(negative values default to 0)

- **AbstractDialogFragment.java**
  - Store `coordinateFormatPosition` in instance variable
  - Save/restore position via `onSaveInstanceState()` /
`onCreate()`
  - Register callback to update position immediately when user
changes format

- **CacheDetailsCreator.java**
  - Enhanced `addCoordinates()` to accept initial position
parameter and return `CoordinatesFormatSwitcher` instance
  - Return type change is backward compatible (only unused by
one caller)
  - Updated JavaDoc to reference correct issue #17418

- **CacheDetailActivity.java**
  - Added coordinate format persistence to `DetailsViewCreator`
inner class for cache coordinates
  - Added coordinate format persistence to
`WaypointsViewCreator` inner class for waypoint coordinates
  - Store per-waypoint format positions in a HashMap (waypoint
ID → format position)
  - Save/restore HashMap directly using `putSerializable()` /
`getSerializable()` with `@SuppressWarnings("unchecked")` for
optimal performance
  - Register callbacks to update positions immediately when user
changes format

- **WaypointViewHolder.java**
```

```
  - Added `setCoordinateFormatPosition()` to set initial
position
  - Added `setOnCoordinateFormatChangedListener()` to register
callback

### Flow

**Cache coordinates:**
User taps coordinates → callback updates
`coordinateFormatPosition` → position update triggers fragment
recreation → `onCreate()` restores position from bundle →
`addCoordinates()` creates switcher with saved position → format
preserved.

**Waypoint coordinates:**
User taps waypoint coordinates → callback updates HashMap entry
for that waypoint ID → view refresh occurs → `fillViewHolder`
retrieves position from HashMap → format preserved per waypoint.

### Coverage

The fix now covers coordinate format persistence in:
- Cache/waypoint info sheets (map popups) - via
`AbstractDialogFragment`
- Cache detail activity (cache coordinates) - via
`CacheDetailActivity.DetailsViewCreator`
- Waypoint list in cache detail - via
`CacheDetailActivity.WaypointsViewCreator`

## Related issues

Related to #17418

## Additional context

Logging added when invalid position values are corrected.
JavaDoc documents API changes and parameter validation behavior.
All implementations use consistent state management patterns.

For waypoints, each waypoint maintains its own format position
independently using a HashMap-based approach. The HashMap is
serialized directly to the Bundle with
`@SuppressWarnings("unchecked")` for the unchecked cast during
deserialization. This approach is preferred over runtime type
checking for performance reasons, as we have full control over
the data being stored and deserialized. The implementation is
simpler (compared to parallel arrays) and more efficient.
```

<issue_title>Coordinate format overwritten in cache/waypoint info sheet</issue_title>
><issue_description>I'm using c:geo for getting the coordinates for my car's navigation system. In contrast to c:geo's `N xx xx.xxx E xxx xx.xxx` default format, my car prefers `N xx xx xx.x E xx xx xx.x` format. This can be easily achieved by a single tap on the cache's/waypoint's coordinates in the info sheet. Only that it gets reset to `N xx xx xx.x E xx xx xx.x` format automatically after a few seconds! Not a big deal, but still annoying, and may be confusing if you don't look closely enough.
>
> Happens on a physical device only, was not able to reproduce this in the emulator.
> Observed with several releases 2025.07.xx and 2025.08.xx</issue_description>
>
><agent_instructions>Try to solve this issue with storing the selected / changed coordinate-format in the bundle to recreate the view correct, as eddiemuc suggested in his last comment. </agent_instructions>
>
> ## Comments on the Issue (you are @copilot in this section)
>
><comments>
><comment_new><author>@murggel</author>
> @moving-bits does this still happens?</comment_new>
><comment_new><author>@SammysHP</author>
> Could this be caused by the position update, which updates the distance info in the way point list?</comment_new>
><comment_new><author>@moving-bits</author>
> > @moving-bits does this still happens?
>
> Will need to retest. But it happens only when you are "on the move", getting position updates seems to be a prerequisite.
>
> > Could this be caused by the position update, which updates the distance info in the way point list?
>
> I observed this error in the cache info sheet on the map only yet, can't say if it happens on cache detail page as well.
> </comment_new>
><comment_new><author>@eddiemuc</author>
> > Could this be caused by the position update, which updates the distance info in the way point list?

```
&gt;
&gt; pretty sure that @SammysHP s idea is correct. Whatever
causes a refresh of the displayed data (probably position
update, but maybe also simply because Android decides to refresh
a view for whatever reason) might reset the format. I assume we
don't store it in the activities bundle or (in case of waypoint-
list) on a per-waypoint-base for usage in (re)creating item
views.</comment_new>
&gt;<comment_new><author>@murggel</author>
&gt; I would like to try to fix this with the copilot-agent to
test / get used to it...</comment_new>
&gt;</comments>
&gt;</pr_description>

## Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>









of the waypoints are set in WaypointsViewCreator.fillViewHolder
</comment_old>

<comment_old>
<author>@murggel</author>
@copilot this PR is related to issue #17418 not to #4789.
</comment_old>

<comment_old>
<author>@copilot</author>

&gt; @copilot For the coordinates of the waypoints in
CacheDetailActivity the format position needs to be persisted as
well. The coordinates of the waypoints are set in
WaypointsViewCreator.fillViewHolder

Added coordinate format persistence for waypoints in
WaypointsViewCreator.fillViewHolder. Each waypoint's format
position is now stored in a Map and persisted via onCreate/
onSaveInstanceState. Updated WaypointViewHolder to support
position management and callbacks. Commit 8bf8ecb.
```

</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot this PR is related to issue #17418 not to #4789.

You're correct. The PR should reference issue #17418, not #4789. The commit messages and PR description reference the wrong issue number. The PR title and linked issue will need to be updated to point to #17418.
</comment_old>

<comment_old>
<author>@murggel</author>
@copilot why do you use one base string (`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two suffices instead of two strings directly?
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot why do you use one base string (`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two suffices instead of two strings directly?

You're right - using two direct strings is simpler and clearer. Changed to `STATE_WAYPOINT_IDS` and `STATE_WAYPOINT_POSITIONS` constants instead of base string + suffixes. Commit 1f65ee6.
</comment_old>

<comment_old>
<author>@murggel</author>
>
> > @copilot why do you use one base string (`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two suffices instead of two strings directly?
>
> You're right - using two direct strings is simpler and clearer. Changed to `STATE_WAYPOINT_IDS` and `STATE_WAYPOINT_POSITIONS` constants instead of base string + suffixes. Commit 1f65ee6.

@copilot can you use a map to save the format-position? The waypoint id would be the key and the format-position the value. Can this map then be saved directly in the bundle?

```
</comment_old>

<comment_old>
<author>@copilot</author>

&gt; &gt;
&gt; &gt; &gt; @copilot why do you use one base string
(`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two suffices
instead of two strings directly?
&gt; &gt; ...

Yes, the Map can be saved directly. Changed to use
`putSerializable()` to save the HashMap directly instead of
converting to parallel int arrays. This simplifies the code
significantly - reduced from ~20 lines to ~6 lines for save/
restore logic. Commit 9a2fabe.
</comment_old>

<comment_old>
<author>@murggel</author>
&gt;
&gt; &gt; &gt;
&gt; &gt; &gt; &gt; @copilot why do you use one base string
(`STATE_WAYPOINT_COORDINATE_FORMAT_POSITIONS`) and two suffices
instead of two strings directly?
&gt; &gt; &gt; ...
&gt;
&gt; Yes, the Map can be saved directly. Changed to use
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converting to parallel int arrays. This simplifies the code
significantly - reduced from ~20 lines to ~6 lines for save/
restore logic. Commit 9a2fabe.

And it does not have to rely on same order of the two int
arrays
</comment_old>

<comment_old>
<author>@murggel</author>
@copilot can you change the code, so that suppressing the
warning is not necessary?
</comment_old>

<comment_old>
<author>@copilot</author>

&gt; @copilot can you change the code, so that suppressing the
```

warning is not necessary?

Removed `@SuppressWarnings("unchecked")` by using runtime type checking with `instanceof`. The code now validates that the deserialized object is a HashMap and that each entry has Integer keys and values before adding them to the map. Commit 33696ba.
</comment_old>

<comment_old>
<author>@murggel</author>
> > @copilot can you change the code, so that suppressing the warning is not necessary?
>
> Removed `@SuppressWarnings("unchecked")` by using runtime type checking with `instanceof`. The code now validates that the deserialized object is a HashMap and that each entry has Integer keys and values before adding them to the map. Commit 33696ba.

@copilot: Does this commit have performance-impact? Since we have the control over the stored type in the bundle, maybe we should go with the suppress-warning-approach? What do you think?
</comment_old>

<comment_old>
<author>@copilot</author>

> > > @copilot can you change the code, so that suppressing the warning is not necessary?
> >
> > Removed `@SuppressWarnings("unchecked")` by using runtime type checking with `instanceof`. The code now v...

You're right about the performance impact. The runtime type checking adds O(n) instanceof operations (1 HashMap check + 2n Integer checks for n waypoints) during each fragment recreation. Since we control what's stored in the Bundle, the `@SuppressWarnings` approach is more appropriate - simpler code and better performance. Reverted to the suppressed warning approach with direct cast and `putAll()`. Commit d438405.
</comment_old>

<comment_old>
<author>@murggel</author>
@copilot the reference issue is #17418 instead of #4789

please change it for all places in comments
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot the reference issue is #17418 instead of #4789
>
> please change it for all places in comments

Already fixed in commit 64a6683. The JavaDoc comment in CacheDetailsCreator.java now references #17418. No other occurrences of #4789 were found in comments across all modified files.
</comment_old>

<comment_new>
<comment_id>3648078906</comment_id>
<author>copilot-pull-request-reviewer[bot]</author>
Pull request overview

This PR fixes coordinate format selection persistence in cache/waypoint info sheets and cache detail activity. When users change the coordinate format by tapping on coordinates, the selection now persists across fragment recreations triggered by position updates.

Changes:
- Added position persistence callbacks to `CoordinatesFormatSwitcher` to notify format changes immediately
- Implemented state management in `AbstractDialogFragment` and `CacheDetailActivity` inner classes to save/restore coordinate format positions
- Enhanced `CacheDetailsCreator.addCoordinates()` to accept initial format position and return the switcher instance for callback registration

Reviewed changes

Copilot reviewed 5 out of 5 changed files in this pull request and generated 1 comment.

</comment_new>

</pr_comments>

<pr_reviews>

<comment_thread_resolved>
<file>main/src/main/java/cgeo/geocaching/ui/
CacheDetailsCreator.java:360</file>
<original_commit>64a6683</original_commit>

<comment_old>
<author>copilot-pull-request-reviewer[bot]</author>
The JavaDoc states that the formatPosition parameter "will be normalized to valid range using modulo", but the actual implementation in CoordinatesFormatSwitcher.setPosition() treats negative values specially by defaulting them to 0 rather than applying modulo. The JavaDoc should be updated to accurately reflect this behavior: negative values default to 0, while non-negative values are normalized using modulo.
```suggestion
     * @param formatPosition the initial format position
(negative values default to 0; non-negative values are
normalized to valid range using modulo)
```
</comment_old>

</comment_thread_resolved>

<comment_thread>
<file>main/src/main/java/cgeo/geocaching/ui/
CoordinatesFormatSwitcher.java:63</file>
<original_commit>81220d5</original_commit>

<comment_new>
<comment_id>2679987202</comment_id>
<author>@murggel</author>
@copilot Is using the listener-interface instead of lambda-callback the better way here?
</comment_new>

</comment_thread>

</pr_reviews>

</comments>

```
----
The last **15** git commits in this branch are the changes you
have made so far. Use those as your change commit history.
</problem_statement>
```

```
Analyze and determine if action or explanation is needed
<rules_for_making_changes>
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<comments> before making any changes. You will only address the
<comment_new> comments. Reply to new comments as you address them.
Ignore comments that are not actionable or directed at you.
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suggestion.
    - Is it addressed to you or someone else, e.g. "Hey
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ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>
```

```
A **CI/Build Failure** has been mentioned in the problem statement,
make sure to follow your instructions for addressing CI/Build
Failures.
```